

SURROUND

Material Carbon Passport

22@ Poblenu Residential Block

Site: 22@ Poblenu, Barcelona, Spain

Building Type: Residential

Stage A1–A3 Embodied Carbon Assessment

1. Material Carbon Passport – Key Metrics

434.43 t	1337 kg/m ²	+457.0%
Total CO ₂ e (tonnes)	Carbon Intensity	vs 240.0 kg/m ² benchmark (above benchmark)

2. Carbon Split by Building Element

Element	CO ₂ e (kg)	CO ₂ e (t)	% of Total
Floor	428,610.0	428.61	98.7%
Wall	4,768.5	4.77	1.1%
Roof	1,047.6	1.05	0.2%
TOTAL	434,426.1	434.43	100%

3. Material Comparative Table – Top Choice per Surface

The table below shows the highest-ranked material option for each building surface, selected by combined score (70% lowest carbon / 30% nearest supplier).

Surface	Material	Product Name	Manufacturer	Country	CO ₂ e (kg)	CO ₂ e/m ²
Floor	Steel	GreenSteel Structural Section (RHS 150x100)	Pacific Steel Manufacturing	AU	428610.0	3571.75
Roof	Wood	StructaBoard Timber CLT Panel	AusTimber Innovations Pty Ltd	AU	1047.6	8.73
Wall	Concrete	EcoSlab Ready-Mix Concrete 32 MPa	Concrete Solutions Pty Ltd	AU	4768.5	56.1

4. Baseline vs Altered Scenario

Comparison of the selected EPD-verified material set against the 22@ Poblenu district average of 207 kg CO₂e/m² and the industry benchmark of 240.0 kg CO₂e/m².

Scenario	Intensity (kg CO ₂ e/m ²)	vs Benchmark
22@ District Average	207	-13.8%
Industry Benchmark	240.0	0.0%
This Design (EPD-verified)	1336.7	+457.0%

Generated by Surround Carbon Pipeline · Stage A1–A3 only · Data sourced from EPD database (2050-materials.com) · Benchmark: 240 kg CO₂e/m² (industry standard)